EXPRIMENT-16

16.) Letter Frequency Attack on Monoalphabetic Substitution Cipher

Aim:

To write a Python program to perform a frequency attack on a monoalphabetic substitution cipher.

PROGRAM:



```
from collections import Counter

def frequency_attack(cipher):

    english = "etaoinshrdlcumwfgypbvkjxqz"

    counts = Counter(ch for ch in cipher.lower()
                     if ch.isalpha())

    cipher_order = [ch for ch, count
                    in counts.most_common()]

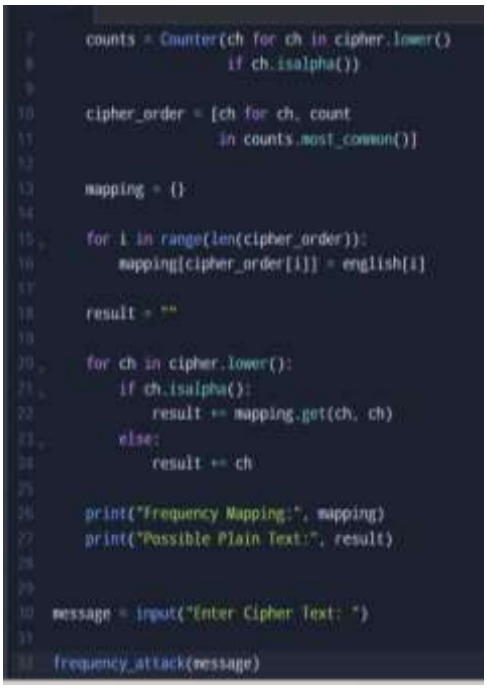
    mapping = {}

    for i in range(len(cipher_order)):
        mapping[cipher_order[i]] = english[i]

    result = ""

    for ch in cipher.lower():
        if ch.isalpha():
            result += mapping.get(ch, ch)
        else:
            result += ch

    print("Frequency Mapping:", mapping)
```



```
    counts = Counter(ch for ch in cipher.lower()
                     if ch.isalpha())

    cipher_order = [ch for ch, count
                    in counts.most_common()]

    mapping = {}

    for i in range(len(cipher_order)):
        mapping[cipher_order[i]] = english[i]

    result = ""

    for ch in cipher.lower():
        if ch.isalpha():
            result += mapping.get(ch, ch)
        else:
            result += ch

    print("Frequency Mapping:", mapping)
    print("Possible Plain Text:", result)


message = input("Enter Cipher Text: ")

frequency_attack(message)
```

OUTPUT:

```
Enter Cipher Text: KHOOR
Frequency Mapping: {'o': 'e', 'k': 't', 'h': 'a', 'r': 'o'}
Possible Plain Text: taeeo
```

RESULT:

Thus, a possible plaintext was estimated using letter frequency analysis.